

Verifying Reasoning Control via Stepwise Introduction of Reward Function Sets

Gemma-3 1B + GRPO (Tunix)

1. Objective and Core Hypothesis

Objective

We gradually evolve the reward design from “correctness reward” to “failure prevention (failure-aware)” and examine how the reasoning accuracy and consistency of a small-scale LLM change.

Hypothesis

“Small models become stronger by ‘prohibiting’ rather than ‘praising.’”

Penalties for failures (constraints) optimize the exploration space more efficiently than rewards for correctness (positive points).

2. Design Philosophy of Reward Sets

Phase	Strategy	Expected Effect
logiconly	Outcome-focused	Judge only whether the final numeric result is correct
grpoft	Format compliance	Enforce preservation of the XML tag structure
steplogic	Thought guidance	Encourage logical connectives and self-corrections
discipline	Behavior constraints	Heavily penalize missing answers, loops, and evasions
full-aware	Logical auditing	Detect unit contradictions and missed calculations in detail

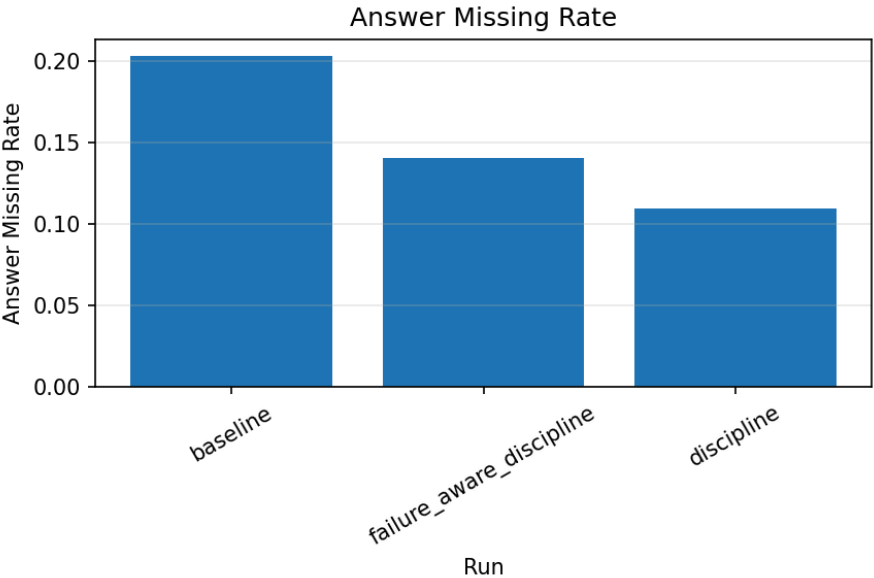
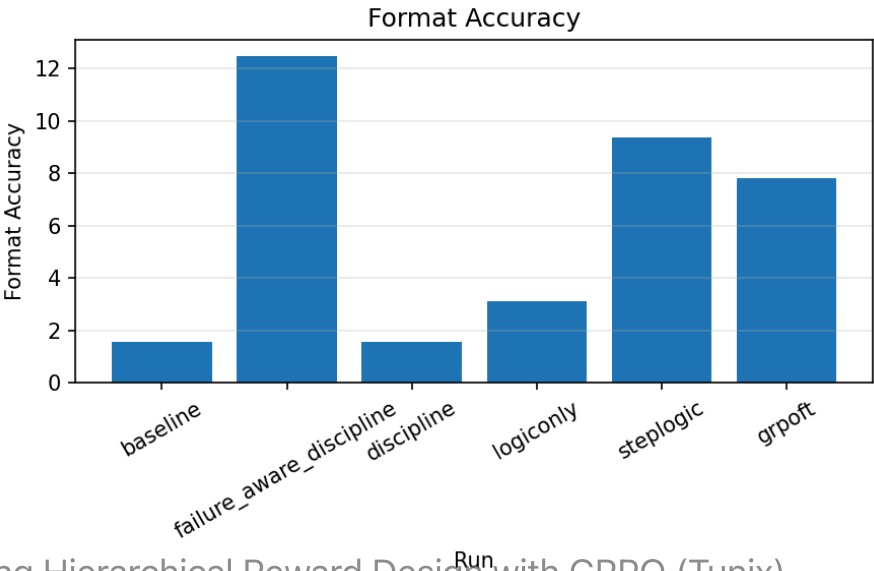
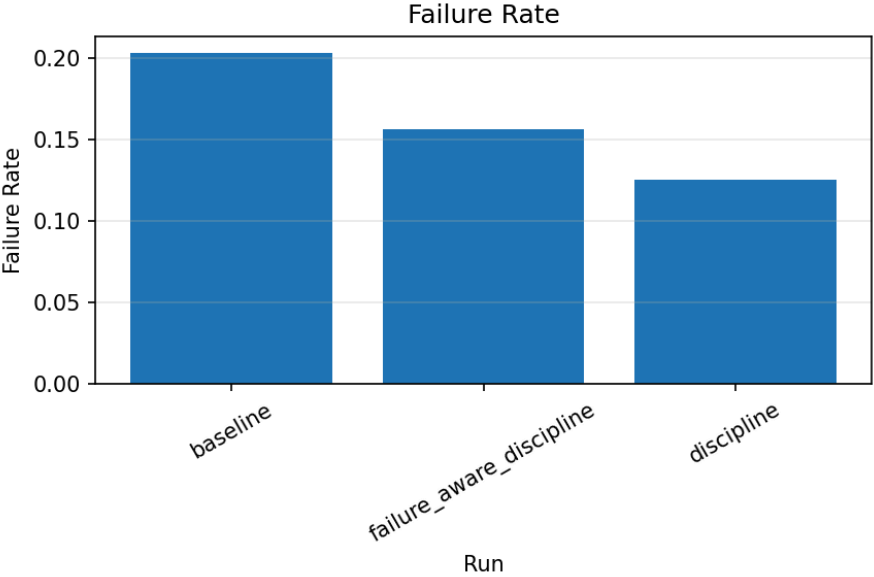
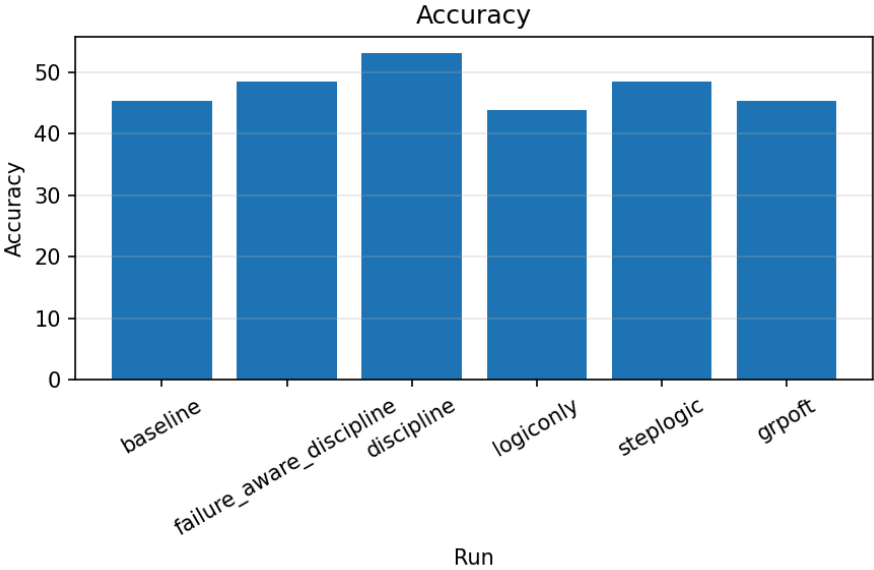
3. Component Functions in Each Set

Set	Key Functions Introduced
1. logiconly	<code>check_answer</code> , <code>check_numbers</code>
2. grpopt	+ <code>match_format</code> (Exactly / Approx)
3. steplogic	+ <code>reward_step_logic</code> , <code>reward_self_correction</code>
4. discipline	+ <code>penalty_failure_like</code> (behavioral discipline)
5. full-aware	+ <code>unit_resolution</code> , <code>stagnation</code> , <code>missing_addends</code>

4. Performance Comparison: Discipline Separates Accuracy

Set	Acc (%)	Format (%)	Failure (%)	Missing (%)
Baseline	45.31	1.56	20.31	20.31
logiconly	43.75	3.13	—	—
grpopt	45.31	7.81	—	—
steplogic	48.44	9.38	—	—
discipline	53.13	1.56	12.50	10.94
full-aware	48.44	12.50	15.63	14.06

The Discipline set maximizes accuracy, while the missing-answer rate is nearly halved.



5. Discussion: Why Did Accuracy Improve?

5.1 Improved Commitment via “Discipline”

- **Cause:** A heavy penalty for missing answers (-3.0) blocks “escape.”
- **Result:** The model terminates ambiguous reasoning earlier and shows improved “responsibility” to output a numeric answer.

5.2 Emergence of Self-Correction

- **Cause:** Positive points for metacognitive words such as “wait.”
- **Result:** Behaviors such as doubting its own reasoning and recomputing become more consistent.

6. Discussion: Side Effects of Over-Constraint

Trade-off Between Format Compliance and Reasoning Accuracy

- **Observation:** In `full-aware`, format compliance is maximized at 12.5%, but accuracy drops.
- **Analysis:**
 - Strict auditing of unit consistency and numeric coverage.
 - For a 1B model, jointly satisfying “format” and “logic” may impose high cognitive load.

7. Conclusion and Future Work

Conclusion

For optimizing small models, **penalties for failures (Discipline)** that restrict the exploration space are extremely effective.

Future Work

- **Curriculum learning:** Gradually strengthen discipline in the later stage of training.
- **Dynamic weighting:** Automatically balance format compliance and reasoning accuracy.

Appendix

- Detailed error analysis (Before/After)

Detailed Error Analysis (Before/After)

Example of a Failed Response

Question:
Jam has three boxes full of pencils and 2 loose pencils which give a total of 26 pencils. If her sister, Meg, has 46 pencils, how many boxes do Jam and Meg need to store all their pencils?

Correct Answer:
9

Response: Reasoning:
First, we need to find the total number of pencils Jam has. We are given that Jam has 26 pencils and 2 loose pencils, so the total number of pencils is $26 + 2 = 28$ pencils. However, the problem states that Jam has 26 pencils. This is a contradiction. Let's assume the problem meant that Jam has 26 pencils.

Next, we need to find the total number of pencils Meg has. We are given that Meg has 46 pencils.

Now, we need to find the total number of pencils they need to store. The total number of pencils is 26 (Jam) + 46 (Meg) = 72 pencils.

Since each box can hold pencils, we need to find how many boxes are needed to store 72 pencils. Let 'x' be the number of boxes. Then, $x * \text{pencils per box} = 72$.

So, $x = 72 / \text{pencils per box}$.

However, we are not given the number of pencils per box. We are given that Jam has 26 pencils and 2 loose pencils, totaling 28 pencils. The problem states that Jam has 26 pencils. We are also given that Meg has 46 pencils. The total number of pencils is $26 + 46 = 72$ pencils.

Let 'b' be the number of boxes. Each box can hold 'p' pencils. Then, $b * p = 72$. We need to find the number of boxes needed to store all the pencils.

We are given that Jam has 26 pencils and 2 loose pencils, for a total of 28 pencils. We are also given that Meg has 46 pencils. The total number of pencils is $26 + 46 = 72$.

We need to find the number of boxes needed to store 72 pencils.

Let 'x' be the number of boxes. Then, $x * p = 72$, where 'p' is the number of pencils per box.

We are not given the number of pencils per box. However, we are told that the total number of pencils is $26 + 2 = 28$. The total number of pencils is $26 + 46 = 72$.

We are given that Jam has 26 pencils and 2 loose pencils, for a total of 28 pencils. We are also given that Meg has 46 pencils.

The total number of pencils is $26 + 46 = 72$.

Let 'n' be the number of boxes. Then, $n * \text{pencils per box} = 72$.

We need to find the number of boxes needed to store all the pencils.

We are given that Jam has 26 pencils and 2 loose pencils, for a total of 28 pencils. We are given that Meg has 46 pencils.

The total number of pencils is $26 + 46 = 72$.

We need to find the number of boxes needed to store 72 pencils.

Let 'x' be the number of boxes. Then, $x * \text{pencils per box} = 72$.

We are not given the number of pencils per box.

Let's assume that each box holds 1 pencil. Then, $x = 72$. So, 72 boxes are needed.

However, we are given that Jam has 26 pencils and 2 loose pencils, for a total of 28 pencils. We are given that Meg has 46 pencils.

Typical “Logical Collapse” Pattern

Problem: Three boxes and 2 loose pencils total 26 pencils. Her sister has 46 pencils. How many boxes are needed to store all pencils?

Model Output: (After a reasoning loop, no final answer)

✗ Four Critical Errors

1. **Context misread:** Misinterprets “3 boxes + 2 pencils = 26” as “26 + 2 = 28” and contradicts itself.
2. **Unjustified assumption:** Declares it “unknown” despite sufficient information, and assumes “1 box = 1 pencil.”
3. **Reasoning loop:** Repeats the same equations and paraphrases, with no information gain.
4. **Answer abandonment:** Ends reasoning without producing a conclusion (`<answer>`).

Why Existing Rewards Cannot Prevent This

Blind Spots in the Current Reward System

- It checks the “attitude” but not the “substance.”
 - If the format is correct, it can receive positive points even with a misread.
 - Since looping is not penalized, writing long text becomes justified.
 - It cannot forbid the “leap” of introducing arbitrary assumptions (Assume).

Conclusion: Increasing correctness rewards alone is insufficient.
Penalties that directly target “failure patterns” are necessary.

Proposed Reward Additions (Final)

- **penalize_unjustified_assumptions** : Strong penalty for arbitrary assumptions such as "Assume" and "Suppose."
- **penalize_false_unknowns** : Heavily penalize escaping with "Unknown" when the numbers exist in the prompt.
- **penalize_reasoning_stagnation** : Parse by sentences and penalize stagnation loops with no information gain (no new numbers/equations).
- **reward_unit_resolution** : Reward correct operations (division/inversion) when unit cues like "box" and "per" appear.